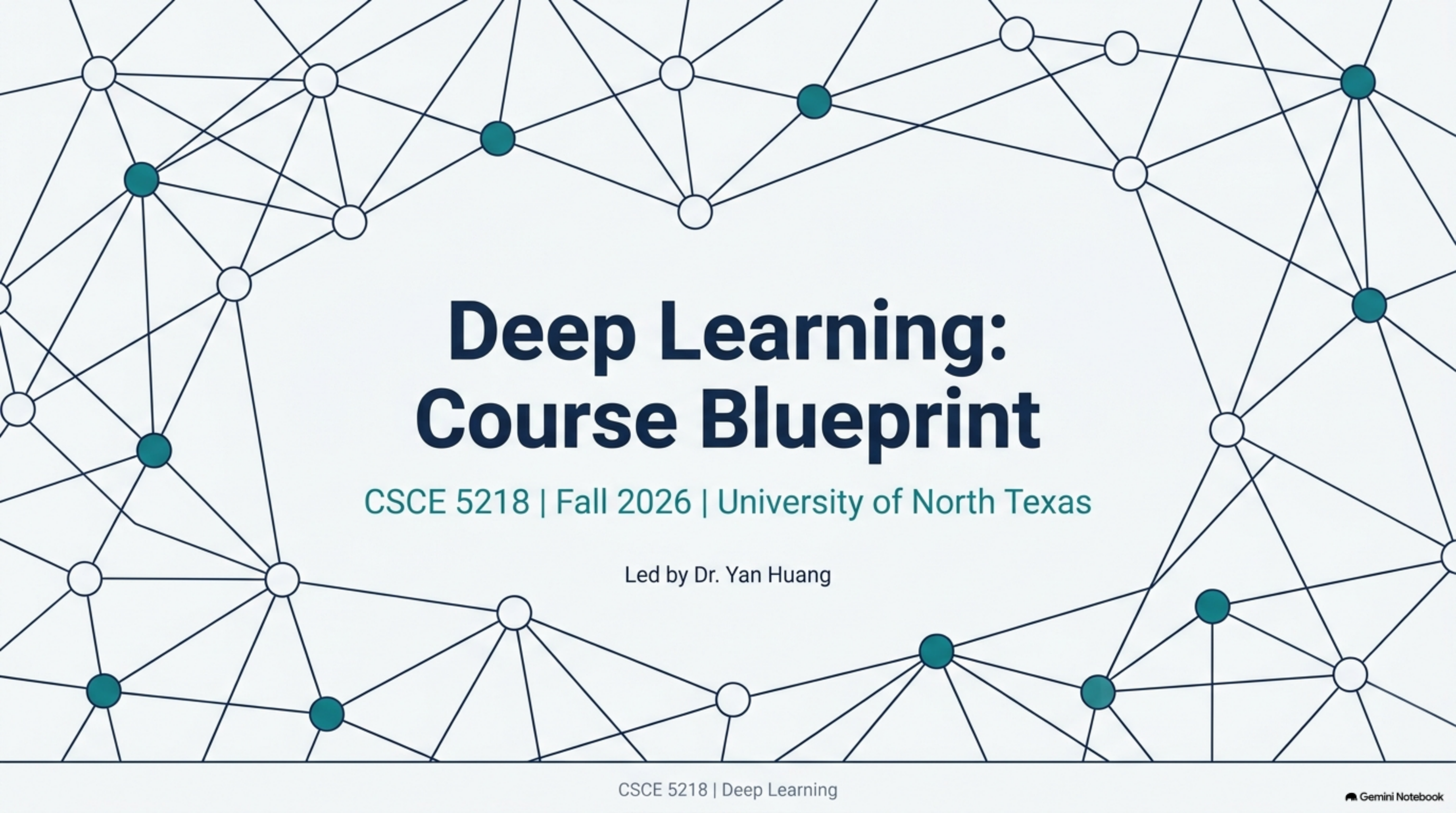




Deep Learning: Course Blueprint

CSCE 5218 | Fall 2026 | University of North Texas

Led by Dr. Yan Huang

Session Details		Instructional Team	
	Thursdays, 12:15–1:35 PM		Instructor: Dr. Yan Huang Yan.Huang@unt.edu Office: F281 Hours: Thu 11:00 AM–12:00 PM
	NTDP D201		Teaching Assistant: Thanh Le ThanhLe5@my.unt.edu Office: F205 Hours: Thu & Fri 4:00–5:00 PM
	Canvas (unt.instructure.com) for materials, assignments, and grades.		
	Expectation: Email is the primary contact method. Expected SLA is a 1-business-day reply. Questions in class and office hours are highly encouraged.		

Deep Learning Readiness

```
graph TD; A[Deep Learning Readiness] --> B[Mathematics]; A --> C[Computer Science]; A --> D[Programming]; B --> B1[Calculus & Linear Algebra.]; C --> C1[Data Structures & Algorithms.]; C --> C2[Must be ready to learn and use standard software development practices (e.g., GitHub).]; D --> D1[Independent Python proficiency (or the ability to learn quickly).]; E[Bonus: Prior Machine Learning background is highly beneficial, though not strictly required.];
```

Mathematics

Calculus & Linear Algebra.

Computer Science

Data Structures & Algorithms.

Must be ready to learn and use standard software development practices (e.g., GitHub).

Programming

Independent Python proficiency (or the ability to learn quickly).

Bonus: Prior Machine Learning background is highly beneficial, though not strictly required.

Dynamic Course Materials



There is no strictly required textbook. Reading materials will be provided dynamically as topics are covered.



Understanding Deep Learning

(Simon J. D. Prince, MIT Press, 2023)



Dive into Deep Learning

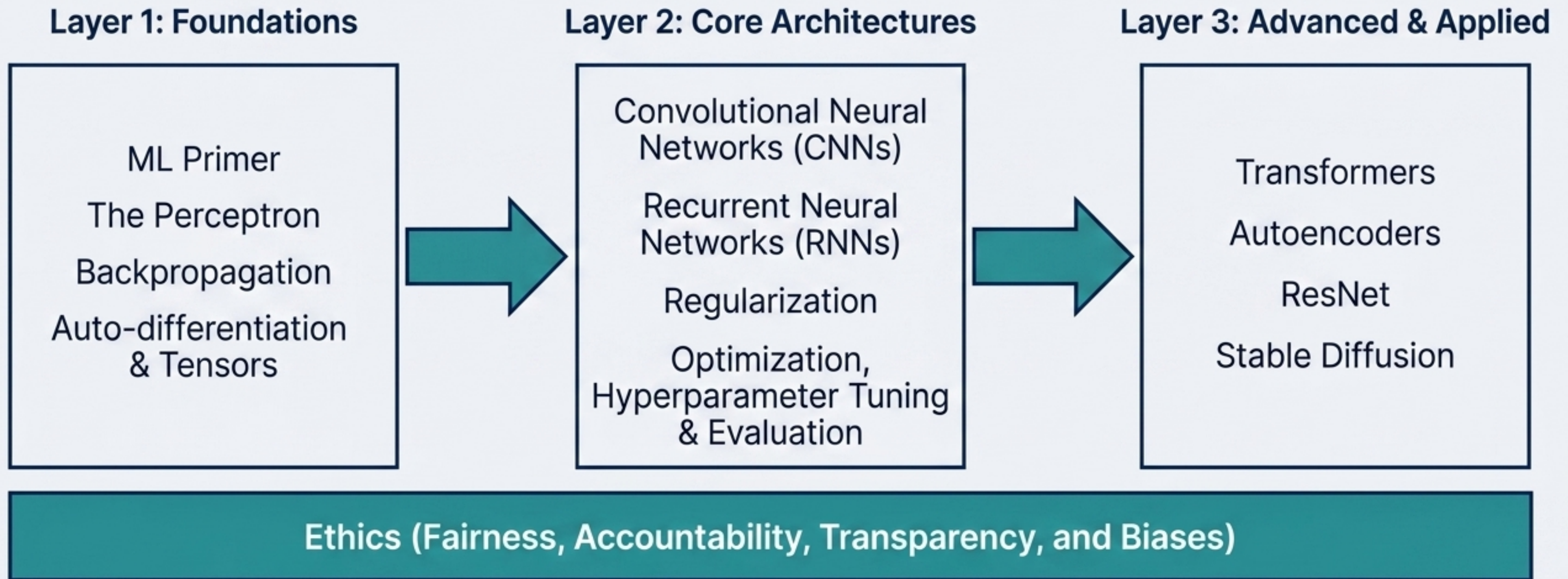
(Zhang, Lipton, Li, & Smola, Cambridge Univ. Press, 2023 | d2l.ai)



Deep Learning: Foundations and Concepts

(C.M. Bishop & H. Bishop, Springer, 2024)

The Learning Pipeline



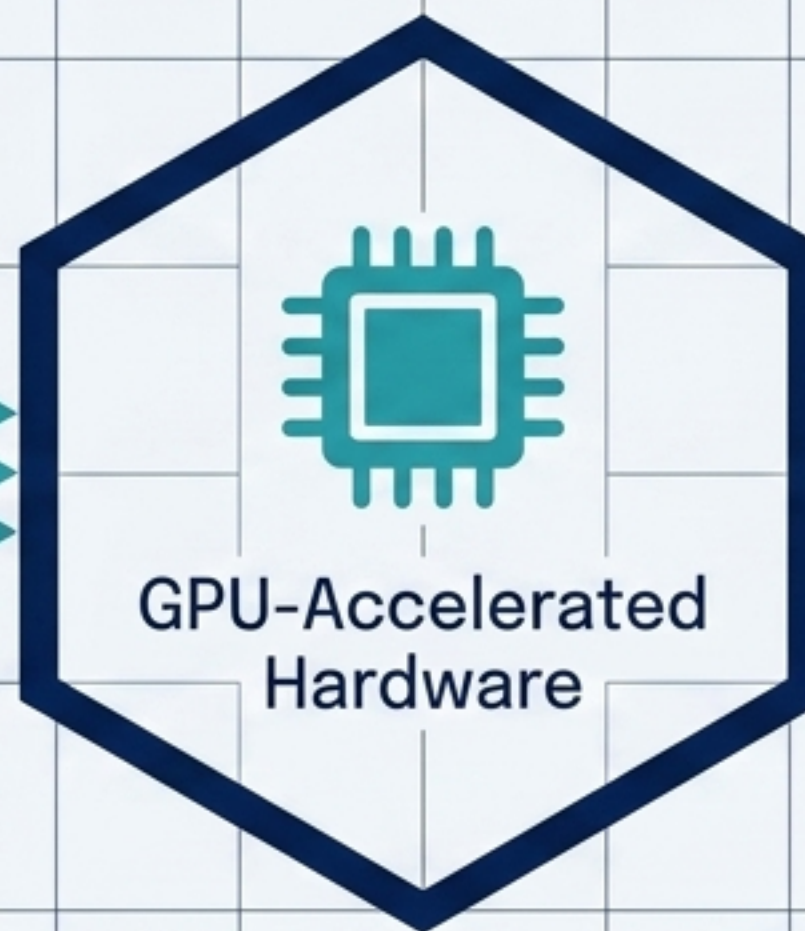
Applied Processing Infrastructure

Raw Input Signals

Speech

Video

Rich
Datasets



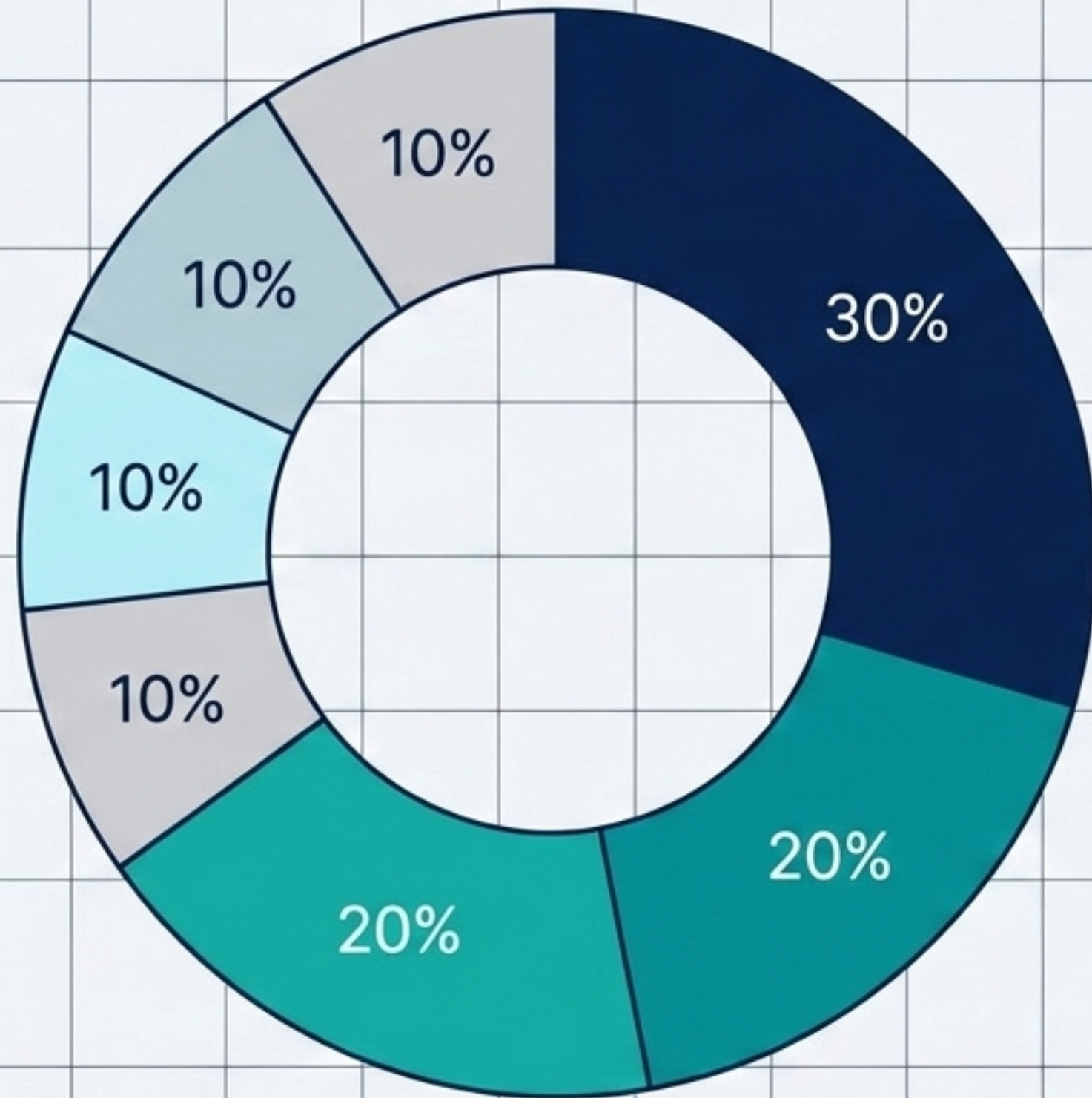
Core Engine: Training multilayer
machine learning models
directly on raw input.

Tool Stack

TensorFlow

PyTorch

Keras

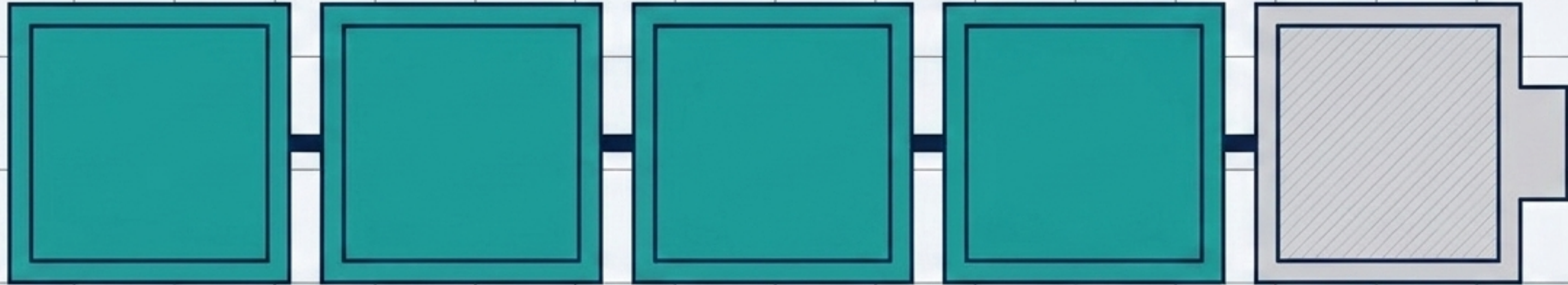


Course Optimization Weights

- Final Exam: 30%
- Course Project: 20%
- Midterm: 20%
- Pre-class Assignments: 10%
- In-class Quizzes: 10%
- Homework Assignments: 10%

Core assessments
(Exams + Project)
account for 70%
of the final
optimization weight.

The Grace Day Buffer



The Resource

5 self-granted extensions available for the entire semester.

Application

Use for extra time on homework assignments without grade penalty.

Strict Limitations

- Cannot be applied to the Course Project.
- Once the 5 days are exhausted, zero late submissions will be accepted.

The Copilot Matrix

Core Principle: AI is a learning and productivity aid, not a replacement for original thinking. You must be able to explain and modify everything you submit.

✓ Acceptable	✗ Unacceptable
- Clarifying concepts - Debugging code - Generating boilerplate - Improving writing clarity	- Generating entire solutions or derivations - Bypassing original thought - Use during quizzes or exams



Mandatory Protocol: All AI assistance requires a brief written disclosure in the submission (e.g., 'Used AI to debug tensor shape errors').

Academic Integrity

(UNT Policy 06.003)

- Collaboration on general problem-solving approaches is permitted.
- Individual code/solution submission is strictly mandatory unless explicitly stated otherwise.
- Plagiarism/cheating from any source (internet, peers) triggers severe university sanctions.

Attendance Protocol

- Expected regularly and randomly checked.
- Proactive communication with the instructional team is required for any absence.
- Illness policy: Stay home to protect community health, but notify the professor immediately to mitigate impact.

Inclusive Framework

ADA Policy

Reasonable accommodations provided. Register with the Office of Disability Accommodation (ODA) and deliver notices to faculty as early as possible.

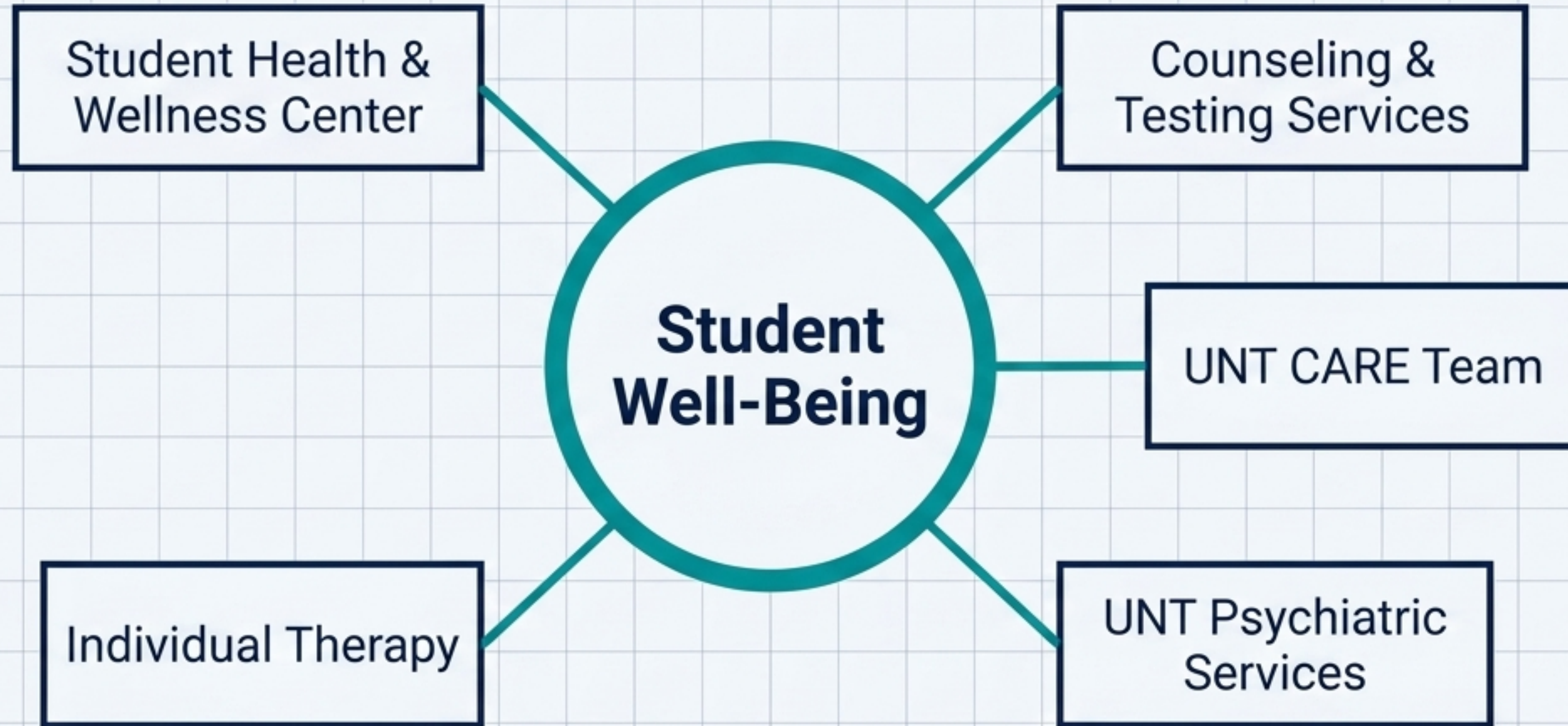
Non-Discrimination

(Policy 16.004) Strict prohibition of harassment or discrimination based on race, color, national origin, religion, sex, sexual orientation, gender identity/expression, age, disability, genetic info, or veteran status.

Religious Observance

No-penalty accommodations for recognized holy days (per Tax Code 11.20). Requires proactive early notification to the instructor.

Campus Support Infrastructure



Life-threatening emergencies: Dial 911 | Mental Health Crisis Line: 940-565-2741 (Press 1)

Initialization Sequence

Start Node



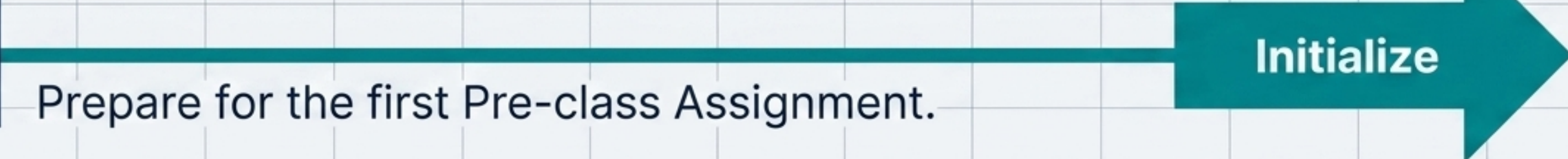
Log into Canvas (unt.instructure.com) to verify access.



Review prerequisite requirements and ensure your Python/GitHub environments are functional.



Prepare for the first Pre-class Assignment.



Initialize

See you in NTDP D201
on Thursday.